

Nexora Corporate Knowledge Base: Architecture Standards

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1. Architectural Paradigm

Nexora systems follow a decoupled, microservices architecture. Core backend services are written in Go for high-throughput concurrency, machine learning & analytics pipelines in Python 3.11+, and frontend web applications in React 19 / TypeScript with Vite.

2. API Gateway & Communication

External and frontend-to-backend communication uses GraphQL and REST over HTTPS with strict OpenAPI / JSON schema validation. Inter-service backend communication uses gRPC with Protocol Buffers and Kafka message streams for asynchronous event-driven workflows.

3. Database & Caching Standards

Transactional persistence uses PostgreSQL 16 with read replicas and connection pooling via PgBouncer. Caching layers utilize Redis clusters with defined TTL policies. Vector search is powered by ChromaDB / pgvector.

4. Infrastructure & Container Orchestration

All workloads run as Docker containers orchestrated via Kubernetes (EKS/GKE). Infrastructure is provisioned strictly via Terraform (Infrastructure as Code).